EC301 — Electronic Devices

Semester III • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Opto-Electronics & IC Fabrication Process

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Optical Absorption in Semiconductors

Optoelectronic devices operate via the conversion of optical energy (photons) into electrical energy (electrons and holes), or vice versa.

1.1 Fundamental Optical Absorption Process

When incident photons strike a semiconductor:

- If photon energy is less than the bandgap ($h\nu < E_g$), the semiconductor cannot absorb the photon; the material is transparent to that wavelength.
- If photon energy equals or exceeds the bandgap ($h\nu \geq E_g$), the photon is absorbed, elevating a valence electron into the conduction band and creating an **electron-hole pair (EHP)**.
- The **cut-off wavelength** (λ_c) above which no band-to-band optical absorption can occur is:

$$\lambda_c = \frac{hc}{E_g} = \frac{1.24}{E_g\,(\text{eV})}\,\mu\text{m} \tag{1}$$

For Silicon ($E_g = 1.12\,\text{eV}$), $\lambda_c \approx 1.11\,\mu$m (near infrared). For GaAs ($E_g = 1.42\,\text{eV}$), $\lambda_c \approx 0.87\,\mu$m.

1.2 Absorption Coefficient and Beer-Lambert's Law

The attenuation of optical intensity $I(x)$ as light propagates through a distance x within a semiconductor is described by the **Beer-Lambert Law**:

Beer-Lambert Absorption Law

$$I(x) = I_0 \exp(-\alpha x) \tag{2}$$

where I_0 is the incident light intensity (after surface reflection), and α is the **absorption coefficient** (cm^{-1}). The characteristic **optical penetration depth** is $\delta = 1/\alpha$. Direct bandgap semiconductors exhibit very steep absorption edges ($\alpha \sim 10^4$–$10^5\,\text{cm}^{-1}$, $\delta \sim 0.1$–$1\,\mu$m), whereas indirect bandgap semiconductors require phonon assistance ($\alpha \sim 10^2$–$10^3\,\text{cm}^{-1}$, $\delta \sim 10$–$100\,\mu$m).

2. Photovoltaic Effects and Solar Cells

A **solar cell** is an un-biased, large-area p-n junction that converts optical solar radiation directly into electrical power via the **photovoltaic effect**.

2.1 Physical Working Principle

1. Photons with $h\nu \geq E_g$ penetrate the thin top layer and generate EHPs within and adjacent to the space charge depletion layer.
2. The high built-in electric field $\vec{\mathcal{E}}$ in the depletion region sweeps photogenerated electrons to the n-side and photogenerated holes to the p-side before they can recombine.

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3. This accumulation of charges establishes a forward photovoltage across the terminals, driving an external load current.

2.2 Equivalent Circuit and Solar Cell I-V Characteristics

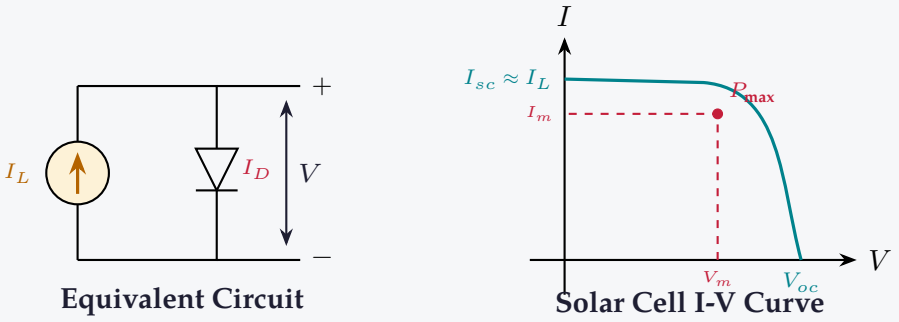
An ideal solar cell is modeled as an ideal light-generated constant current source I_L in parallel with a forward-biased diode:

Solar Cell Current-Voltage Equation

$$I = I_L - I_D = I_L - I_0 \left[\exp\left(\frac{qV}{\eta kT}\right) - 1\right] \tag{3}$$

where I_L is the photogenerated current ($I_L \propto$ incident optical power).

Solar Cell Equivalent Circuit and Fourth-Quadrant I-V Curve



2.3 Key Solar Cell Figures of Merit

- **Short-Circuit Current (I_{sc}):** Current flowing when terminals are shorted ($V = 0$):

$$I_{sc} = I_L \tag{4}$$

- **Open-Circuit Voltage (V_{oc}):** Voltage when terminals are open ($I = 0$):

$$0 = I_L - I_0 \left[\exp\left(\frac{qV_{oc}}{\eta kT}\right) - 1\right] \implies V_{oc} = \eta V_T \ln\left(\frac{I_L}{I_0} + 1\right) \tag{5}$$

- **Maximum Power Point (P_{max}):** Occurs at the knee of the curve where the product $V \cdot I$ is maximized:

$$P_{\text{max}} = V_m I_m \tag{6}$$

- **Fill Factor (FF):** The ratio of maximum usable power to the theoretical maximum product $V_{oc} I_{sc}$:

$$FF = \frac{V_m I_m}{V_{oc} I_{sc}} \quad \text{(typically 0.70–0.85)} \tag{7}$$

- **Power Conversion Efficiency (η):**

$$\eta = \frac{P_{\text{max}}}{P_{\text{in}}} = \frac{V_{oc} I_{sc} FF}{P_{\text{in}}} \tag{8}$$

where P_{in} is the incident solar optical power ($100\,\text{mW/cm}^2$ under AM1.5 standard illumination).

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3. Photodetectors: Photoconductors, Photodiodes, PIN and APD

3.1 Photoconductors

A photoconductor is a slab of semiconductor with ohmic contacts at both ends. Light absorption creates excess electron-hole pairs ($\Delta n = \Delta p = g_L \tau$), increasing the material's bulk conductivity:

$$\Delta\sigma = q(\mu_n + \mu_p)\Delta n = q(\mu_n + \mu_p) g_L \tau \tag{9}$$

The **photoconductive gain** (G) is the number of charge carriers passing through the external circuit per absorbed photon:

$$G = \frac{\tau}{t_{\text{transit}}} = \frac{\tau \mu_n \mathcal{E}}{L} = \frac{\tau \mu_n V}{L^2} \tag{10}$$

Because electron lifetime τ can be much longer than transit time t_{transit}, an electron can circulate through the device multiple times before recombining, producing $G \gg 1$.

3.2 P-N Photodiode

A p-n junction operated under **reverse bias**:

- Reverse bias extends the depletion width and creates a high electric field that sweeps photogenerated carriers out rapidly, generating a drift photocurrent:

$$I_{\text{total}} = I_0 \left(e^{-V_R/V_T} - 1\right) - I_{\text{photo}} \approx -(I_0 + I_{\text{photo}}) \tag{11}$$

- **Responsivity** (R)**:**

$$R = \frac{I_{\text{photo}}}{P_{\text{opt}}} = \frac{\eta_{ext} q}{h\nu} = \frac{\eta_{ext} q \lambda}{hc} \quad \text{(A/W)} \tag{12}$$

  where η_{ext} is the external quantum efficiency.

3.3 PIN Photodiode

A PIN photodiode inserts a wide, undoped **intrinsic** (i) semiconductor layer between heavily doped p^+ and n^+ regions (p^+–i–n^+).

- Even under small reverse bias, the intrinsic layer is completely depleted ($W \approx W_i$).
- **Key Advantages over standard p-n photodiodes**:
  1. **High Quantum Efficiency:** The wide W_i provides a large volume for optical absorption, ensuring that nearly all incident photons are converted into drift carriers rather than slow diffusion carriers.
  2. **Low Junction Capacitance:** Because $C_j = \frac{\epsilon_s A}{W_i}$ and W_i is large, capacitance is extremely small (< 1 pF), drastically reducing the RC time constant.
  3. **High Speed / Bandwidth:** Operating speeds reach 10–40 GHz, making PIN diodes standard in fiber-optic communications.

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3.4 Avalanche Photodiode (APD)

The APD is operated under very high reverse bias close to the avalanche breakdown voltage:

- Photo-generated electrons entering the high-field multiplication depletion region acquire sufficient kinetic energy to cause **impact ionization**, knocking valence electrons into the conduction band.
- This secondary carrier multiplication yields an internal current gain M:

$$M = \frac{I_{\text{out}}}{I_{\text{photo}}} \approx \frac{1}{1-(V_R/V_{BR})^n} \quad (M \sim 10\text{–}1000) \tag{13}$$

- APDs provide superior optical sensitivity for detecting ultra-weak light signals, but introduce avalanche excess noise.

3.5 Phototransistor

A phototransistor is a BJT whose reverse-biased collector-base junction is exposed to light. Photogenerated base current I_{photo} acts as base drive, which is multiplied by the transistor current gain:

$$I_C = (\beta + 1) I_{\text{photo}} \tag{14}$$

It provides high optical sensitivity without external amplifiers, but has lower switching speed than PIN photodiodes due to base charge storage.

4. Light Emitting Diodes and Semiconductor Lasers

4.1 Light Emitting Diode (LED)

An LED is a heavily doped, forward-biased **direct bandgap** p-n junction (e.g., GaAs, GaN, InGaN).

- Forward bias injects minority electrons into the p-side and minority holes into the n-side.
- Carriers undergo **spontaneous radiative recombination**, releasing energy as photons:

$$\lambda \approx \frac{hc}{E_g} = \frac{1.24}{E_g\,(\text{eV})}\,\mu\text{m} \tag{15}$$

- Emits **incoherent**, non-directional light across a spectral bandwidth $\Delta\lambda \approx 20\text{–}50\,\text{nm}$.

4.2 Semiconductor Laser Diode

A laser diode produces **coherent, monochromatic, and highly directional** optical radiation.

1. **Three Radiation Processes:**
   - **Stimulated Absorption:** An incoming photon ($h\nu \approx E_g$) is absorbed, exciting an electron from VB to CB.
   - **Spontaneous Emission:** An electron drops spontaneously to the VB, emitting an incoherent photon of random phase and direction.

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- **Stimulated Emission:** An incident photon stimulates an excited electron in the CB to drop to the VB, emitting a second photon having the **exact same energy, phase, polarization, and direction** as the stimulating photon.

2. **Population Inversion (Bernard-Duraffourg Condition):** To achieve net optical gain (stimulated emission exceeding stimulated absorption), the upper CB states must be more populated than lower VB states. This requires heavy forward current injection such that the separation of quasi-Fermi levels exceeds the photon energy:

$$E_g < h\nu < E_{Fn} - E_{Fp} \tag{16}$$

3. **Optical Cavity (Fabry-Perot Resonator):** Cleaved, polished semiconductor crystal facets at both ends act as partially reflecting mirrors, creating an optical feedback cavity that sustains oscillation above the **threshold current** I_{th}.

5. Integrated Circuit Fabrication Technologies

Modern Silicon integrated circuits are manufactured on high-purity single-crystal Silicon wafers using sequential planar processing steps:

5.1 Overview of Unit Fabrication Steps

Process Step	Chemical / Physical Principle	Role in IC Manufacturing
Thermal Oxidation	$Si + O_2 \rightarrow SiO_2$ (Dry) $Si + 2H_2O \rightarrow SiO_2 + 2H_2$ (Wet)	Creates gate dielectric (SiO_2), diffusion barrier masks, and surface passivation.
Photolithography	Optical UV exposure of photoresist polymer through a photomask followed by chemical development.	Transfers micro/nanometer geometric layout patterns onto wafer layers.
Etching	Chemical dissolution (Wet) or plasma/reactive ion bombardment (Dry RIE).	Selectively removes unmasked oxide, polysilicon, or metal layers with high vertical fidelity.
Thermal Diffusion	High-temperature (900–1150°C) thermal dopant transport into silicon.	Forms wells and junction depths; isotropic lateral diffusion occurs.
Ion Implantation	High-energy electrostatic acceleration (10–400 keV) of dopant ions into silicon lattice.	Forms shallow source/drain junctions, threshold voltage adjustments; highly precise dosage.
Chemical Vapor Deposition (CVD)	Gas-phase precursor chemical reaction at wafer surface (LPCVD, PECVD).	Deposits thin dielectric films (Si_3N_4, SiO_2) and polycrystalline silicon gates.
Sputtering (PVD)	High-energy Argon ion bombardment of metal target dislodging atoms onto wafer.	Deposits metal interconnect layers (Aluminum, Copper, Titanium).

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5.2 Photolithography Step-by-Step Sequence

Photolithography is the repetitive master patterning engine of planar processing:

$$\begin{aligned}\text{Wafer Clean} &\rightarrow \text{Barrier Oxide Growth} \rightarrow \text{Photoresist Coating} \rightarrow \text{Soft Bake} \\ &\rightarrow \text{Mask Align \& UV Exposure} \rightarrow \text{Development} \\ &\rightarrow \text{Hard Bake} \rightarrow \text{Chemical / Plasma Etch} \rightarrow \text{Photoresist Stripping}\end{aligned}$$

- **Positive Photoresist:** Exposed regions become soluble in developer solution and wash away. Final pattern matches mask opaque features.
- **Negative Photoresist:** Exposed regions cross-link and become insoluble; unexposed regions wash away.

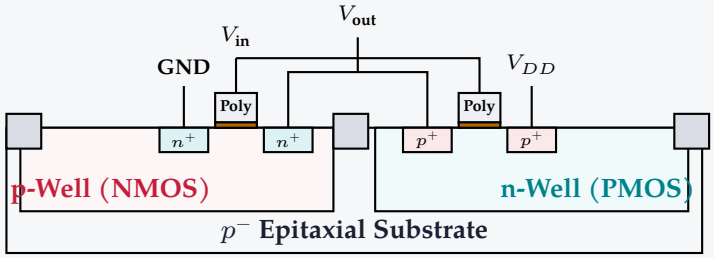
5.3 Twin-Tub CMOS Process Flow

The **twin-tub CMOS process** fabricates both n-channel (NMOS) and p-channel (PMOS) transistors on a single common substrate by creating separate, optimized n-well and p-well tubs:

1. **Substrate Preparation:** Start with a high-resistivity p^- (or n^-) epitaxial silicon wafer.
2. **Tub Formation (Twin-Well):** Implantation of Phosphorus ions to form the n-well (for PMOS) and Boron ions to form the p-well (for NMOS), followed by a high-temperature drive-in diffusion.
3. **Device Isolation (LOCOS or STI):** Shallow Trench Isolation (STI) etched and filled with SiO_2 to isolate adjacent transistor active areas and suppress latch-up.
4. **Threshold Voltage (V_T) Adjustment:** Shallow Boron/Phosphorus channel implants to tune NMOS and PMOS threshold voltages.
5. **Gate Oxide and Polysilicon Gate:** Grow ultra-thin gate oxide ($t_{ox} \sim 1$–$2\,\text{nm}$) followed by LPCVD polysilicon deposition, photolithography, and anisotropic dry etching of gates.
6. **Self-Aligned Source/Drain Implantation:**
   - n^+ Arsenic/Phosphorus implant into p-well using polysilicon gate as mask (forms NMOS source/drain).
   - p^+ Boron implant into n-well using gate as mask (forms PMOS source/drain).
7. **Interlayer Dielectric (ILD) and Contact Holes:** Deposit CVD oxide, pattern contact windows via RIE.
8. **Metallization and Passivation:** Sputter Aluminum/Copper interconnects, pattern metal lines, and deposit protective Si_3N_4 passivation overcoat.

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Cross-Section of Completed Twin-Tub CMOS Inverter



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Quick Revision — Module 4

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Key Formulas and Metrics at a Glance

Optical Absorption Cut-off	$\lambda_c = \frac{hc}{E_g} = \frac{1.24}{E_g\,(\text{eV})}\,\mu\text{m}$
Beer-Lambert Law	$I(x) = I_0 \exp(-\alpha x)$
Solar Cell Open-Circuit Voltage	$V_{oc} = \eta V_T \ln\left(\frac{I_L}{I_0} + 1\right)$
Solar Cell Fill Factor (FF)	$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$
Solar Cell Power Efficiency (η)	$\eta = \frac{V_m I_m}{P_{\text{in}}} = \frac{V_{oc} I_{sc} FF}{P_{\text{in}}}$
Photoconductive Gain (G)	$G = \frac{\tau}{t_{\text{transit}}} = \frac{\tau \mu V}{L^2}$
Photodiode Responsivity (R)	$R = \frac{I_{\text{photo}}}{P_{\text{opt}}} = \frac{\eta_{ext} q \lambda}{hc}$
Phototransistor Collector Current	$I_C = (\beta + 1) I_{\text{photo}}$
Laser Population Inversion	$E_{Fn} - E_{Fp} > h\nu > E_g$
APD Multiplication Factor (M)	$M = \frac{1}{1-(V_R/V_{BR})^n}$

One-Line Definitions

- **Photovoltaic Effect:** Direct generation of electric potential across a p-n junction caused by absorption of incident photons.
- **Fill Factor (FF):** Figure of merit measuring the squareness of a solar cell's fourth-quadrant I-V characteristic.
- **PIN Photodiode:** High-speed, high-efficiency photodetector incorporating a wide intrinsic layer between p^+ and n^+ regions.
- **Avalanche Photodiode (APD):** Highly sensitive photodetector employing internal carrier multiplication via impact ionization.
- **Population Inversion:** Non-equilibrium condition where the number of electrons in higher energy states exceeds that in lower states.
- **Photolithography:** Microfabrication process transferring photographic geometrical circuit patterns onto a photosensitive polymer resist.
- **Dry Reactive Ion Etching (RIE):** High-resolution anisotropic material removal combining chemical reactivity with directional ion bombardment.
- **Chemical Vapor Deposition (CVD):** Deposition of high-purity solid dielectric or polysilicon thin films from vapor-phase chemical precursors.

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Frequently Asked / Viva Questions

Q1. Why does a PIN photodiode possess significantly higher speed and bandwidth than a standard p-n photodiode?

In a standard p-n diode, many carriers are generated in neutral regions and must slowly diffuse to the junction ($t_{\text{diff}} \sim \mu$s). In a PIN diode, the wide intrinsic layer (W_i) ensures nearly all photons are absorbed directly within the depletion field, where carriers drift at saturation velocity ($t_{\text{drift}} \sim$ ps). Furthermore, the wide layer significantly reduces junction capacitance ($C_j = \epsilon A/W_i$), decreasing the RC time constant.

Q2. What is the difference between spontaneous emission and stimulated emission?

Spontaneous emission occurs randomly without external influence, yielding photons of arbitrary direction and incoherent phase (LED operation). Stimulated emission is triggered by an incoming photon, causing an excited electron to drop and emit a twin photon with identical wavelength, phase, direction, and polarization (Laser operation).

Q3. State the necessary conditions for semiconductor laser operation.

- Population inversion across the bandgap ($E_{Fn} - E_{Fp} > h\nu > E_g$).
- Optical cavity feedback (Fabry-Perot resonator) to sustain stimulated emission.
- Injection current density exceeding the threshold current density ($J > J_{th}$) where optical gain exceeds cavity losses.

Q4. What is the physical meaning of the Solar Cell Fill Factor (FF)?

It is the ratio of maximum achievable electric power ($V_m I_m$) to the ideal product ($V_{oc} I_{sc}$). A higher fill factor (approaching 0.85) represents lower internal series resistance ($R_s \to 0$) and higher shunt resistance ($R_{sh} \to \infty$).

Q5. Why is dry oxidation preferred over wet oxidation for gate oxide growth?

Dry oxidation ($\text{Si} + \text{O}_2 \to \text{SiO}_2$) grows very slowly, producing a dense, highly uniform, defect-free oxide film with extremely low interface trap density, which is critical for thin gate dielectrics. Wet oxidation ($\text{Si} + 2\text{H}_2\text{O} \to \text{SiO}_2 + 2\text{H}_2$) is much faster but yields a less dense oxide used for thick masking and field isolation.

Q6. Compare Ion Implantation with Thermal Diffusion for junction doping.

Ion implantation allows independent, ultra-precise control over dopant dose and penetration depth, operates at low temperatures, and avoids lateral diffusion under mask edges. Thermal diffusion requires high temperatures ($> 1000^\circ$C), suffers from lateral diffusion, but causes no crystal lattice damage.

Q7. What is the difference between isotropic and anisotropic etching?

Isotropic etching (wet chemical) etches equally in all directions, causing significant mask undercut. Anisotropic etching (dry plasma / RIE) etches preferentially in the vertical direction, producing vertical sidewalls essential for sub-micron VLSI layouts.

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Frequently Asked / Viva Questions (contd.)

Q8. Why is the twin-tub CMOS process preferred over single-well CMOS?

In single-well CMOS, only one transistor type's well doping can be optimized; the other must use the substrate doping. In the twin-tub process, both n-well and p-well doping profiles and depths can be optimized independently, allowing precise matching of NMOS and PMOS threshold voltages and minimizing latch-up.

Q9. What is photoconductive gain and how can it exceed unity?

$G = \tau / t_{\text{transit}}$. When the minority carrier lifetime τ exceeds the electron transit time t_{transit}, an electron sweeps out of the positive contact while an electron is simultaneously injected from the negative contact to maintain neutrality. The electron circulates through the external circuit multiple times before recombination occurs.

Q10. What is the role of self-alignment in polysilicon gate MOS fabrication?

The polysilicon gate electrode acts as an implantation mask for the source and drain implants. This ensures that the edges of the source and drain line up with the edges of the gate, eliminating parasitic gate-to-source and gate-to-drain overlap capacitances.

Comparison of LED vs. Semiconductor Laser Diode

Parameter	Light Emitting Diode (LED)	Semiconductor Laser Diode
Emission Mechanism	Spontaneous radiative recombination	Stimulated emission with optical cavity feedback
Coherence	Incoherent (random phases)	Highly coherent (identical phase and frequency)
Spectral Width ($\Delta\lambda$)	Wide ($\Delta\lambda \sim 25$–$50\,\text{nm}$)	Extremely narrow ($\Delta\lambda \sim 0.1$–$2\,\text{nm}$, monochromatic)
Beam Divergence	Wide, non-directional (~ 60–120°)	Narrow, highly directional (~ 5–15°)
Threshold Current	No threshold current (emits light at any forward current)	Requires injection current exceeding threshold I_{th}
Modulation Bandwidth	Moderate (tens to hundreds of MHz)	Ultra-high (tens of GHz)
Coupling Efficiency	Low coupling efficiency into optical fibers	High coupling efficiency ($> 80\%$) into single-mode fibers
Applications	Indicators, lighting, short-range plastic fibers	High-speed long-haul optical fiber communications, LiDAR

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